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SEC: B

BATCH: B4

ROLL NO: 47

ML LAB : Practical 3

In [6]:

```
import pandas as pd
import numpy as np
df=pd.read_csv('enjoy.csv')
```

In [7]:

```
X=df.iloc[:,:-1].values
y=df.iloc[:,-1].values
```

In [10]:

```
def find_s(X,y):
    hypothesis=None
    print('initial hypothesis: ',hypothesis)

    for i in range (len(X)):
        if y[i]=='Yes':
            if hypothesis is None:
                hypothesis =X[i].copy()
            else:
                for j in range(len(hypothesis)):
                    if hypothesis[j] !=X[i][j]:
                        hypothesis[j]="?"
        print(f"\after training example {i+1}")
        print(hypothesis)
    return hypothesis
final_hypothesis = find_s(X,y)
print("\n final hypothesis ")
print(final_hypothesis)
```

```
initial hypothesis:  None
⍰fter training example 1
['Sunny' 'Warm' 'Normal' 'Strong' 'Warm' 'Same']
⍰fter training example 2
['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
⍰fter training example 3
['Sunny' 'Warm' '?' 'Strong' 'Warm' 'Same']
⍰fter training example 4
['Sunny' 'Warm' '?' 'Strong' '?' '?']
⍰fter training example 5
['Sunny' 'Warm' '?' 'Strong' '?' '?']
⍰fter training example 6
['Sunny' 'Warm' '?' 'Strong' '?' '?']
⍰fter training example 7
['Sunny' 'Warm' '?' '?' '?' '?']
⍰fter training example 8
['Sunny' 'Warm' '?' '?' '?' '?']
⍰fter training example 9
['Sunny' 'Warm' '?' '?' '?' '?']
⍰fter training example 10
['Sunny' 'Warm' '?' '?' '?' '?']

 final hypothesis
['Sunny' 'Warm' '?' '?' '?' '?']
```

In [18]:

```
def candidate_elimination(concepts, target):
    specific = concepts[0].copy()
    general = [["?" for _ in range(len(specific))]]

    print("\n=== Candidate Elimination Algorithm ===\n")
    print("Initial S:", specific)
    print("Initial G:", general)
    print("\n---------------------------------------\n")

    for i, h in enumerate(concepts):
        if target[i] == 'Yes':  # Positive example
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    specific[x] = "?"
        else:  # Negative example
            for x in range(len(specific)):
                if h[x] != specific[x]:
                    general.append(["?" if j != x else specific[x] for j in range(len(specific))])

        print(f"Example {i+1}:")
        print("S =", specific)
        print("G =", general)
        print("\n---------------------------------------\n")

    print("\n=== Final Hypotheses ===")
    print("Specific (S):", specific)
    print("General (G):", general)
    print("\n========================\n")

    return specific, general


S, G = candidate_elimination(X, y)
```

```
=== Candidate Elimination Algorithm ===

Initial S: ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
Initial G: [['?', '?', '?', '?', '?', '?']]

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Example 1:
S = ['Sunny', 'Warm', 'Normal', 'Strong', 'Warm', 'Same']
G = [['?', '?', '?', '?', '?', '?']]

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Example 2:
S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G = [['?', '?', '?', '?', '?', '?']]

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Example 3:
S = ['Sunny', 'Warm', '?', 'Strong', 'Warm', 'Same']
G = [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
'?', '?', '?', 'Same']]

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Example 4:
S = ['Sunny', 'Warm', '?', 'Strong', '?', '?']
G = [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?',
'?', '?', '?', 'Same']]

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=== Final Hypotheses ===
Specific (S): ['Sunny', 'Warm', '?', 'Strong', '?', '?']
General (G): [['?', '?', '?', '?', '?', '?'], ['Sunny', '?', '?', '?', '?', '?'], ['?', 'Warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'
['?', '?', '?', '?', '?', 'Same']]
```

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In [ ]:
```